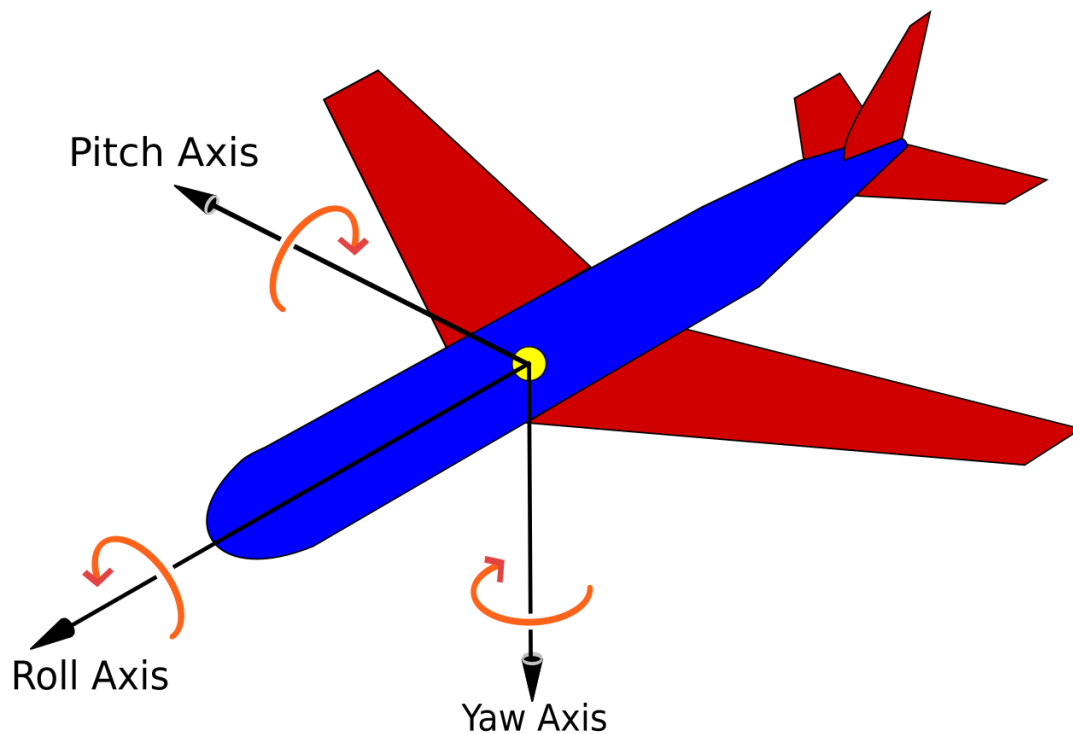


MPU6050 3D Aircraft Orientation Fix Guide

Edit only processing app code. (Plane_Sim_3D.pde)

This guide helps you fix incorrect aircraft movement by editing only one line of code.

Orientation Fix.



Orientation Fix (Very Simple Guide)

If your airplane is moving in the **wrong direction**, you only need to edit **ONE line** in the code.

Find This Line

```
applyMatrix(getRotationMatrix(yaw, pitch+45, roll));
```

 This is the **ONLY line you need to change**

Common Fixes

1. Plane is tilted when sensor is flat

👉 Change:

pitch + 45 → pitch

✓ Result:

```
applyMatrix(getRotationMatrix(yaw, pitch, roll));
```

2. Pitch is reversed (forward/backward wrong)

👉 Change:

pitch → -pitch

✓ Result:

```
applyMatrix(getRotationMatrix(yaw, -pitch, roll));
```

3. Roll is reversed (left/right tilt wrong)

👉 Change:

roll → -roll

✓ Result:

```
applyMatrix(getRotationMatrix(yaw, pitch, -roll));
```

4. Yaw is reversed (turning wrong direction)

👉 Change:

yaw → -yaw

✓ Result:

```
applyMatrix(getRotationMatrix(-yaw, pitch, roll));
```

5. Movement completely wrong (axes mixed)

👉 Try swapping values:

```
applyMatrix(getRotationMatrix(yaw, roll, pitch));
```

Simple Rule

- Wrong forward/back movement → fix **pitch**
- Wrong left/right tilt → fix **roll**
- Wrong turning direction → fix **yaw**

Fine Adjustment

If the plane is slightly off:

pitch + 5

pitch - 10

 Adjust until the plane is level when the sensor is flat.

Important

 **Do NOT change anything else in the code**

Only edit this line:

```
applyMatrix(getRotationMatrix(...));
```

Result

- ✓ Works with any MPU6050 orientation
- ✓ No coding knowledge required
- ✓ Quick trial-and-error fix